

IoT281 Project

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Introduction

The Internet of Things (IoT) refers to a network of physical devices that contain technologies such as sensors, software and network connectivity, allowing them to collect and share data (IBM, 2023). IoT systems can collect information from their environment and use this information to make decisions or respond to different inputs; one example of this is a smart smoke detector, which can use sensors to detect smoke and give an alert when necessary (Amazon Web Services, 2022).

Background Knowledge

Arduino Uno

The Arduino Uno is a microcontroller board based on the ATmega328P microcontroller (ARDUINO, 2021). It has 14 digital input/output pins and 6 analogue input pins, which allow it to communicate with different electronic components (ARDUINO, 2021). Waseem (2026) states that the analogue inputs use a 10-bit ADC, allowing readings between 0 and 1023. In our project, the Arduino Uno acts as the main controller by receiving the MQ-2 sensor reading and controlling the LEDs, buzzer and LCD according to the programmed conditions. The Arduino Uno is suitable for this project because it provides both analogue inputs for the sensor and digital outputs for the system's other components.

MQ-2 Smoke/Gas Sensor

The MQ-2 is a gas sensor that can be used to detect gases such as hydrogen, LPG, methane, carbon monoxide, alcohol, smoke and propane (Seeed Studio, n.d.). According to Seeed Studio (n.d.), the sensor's output voltage increases as the concentration of gas increases, allowing changes in gas levels to be detected. The sensitivity of the MQ-2 can also be adjusted using a potentiometer (Seeed Studio, n.d.). In our project, the MQ-2 is used to detect smoke and provide input to the Arduino Uno. The Arduino can then use the sensor reading to determine whether the system should remain in a safe state or activate the alarm.

16×2 LCD Display

A 16 x2 LCD is a display that can show 16 characters on each of its two lines, allowing up to 32 characters to be displayed at a time (Raj, 2015). According to Söderby (2021), LCDs can be connected to an Arduino to display information such as sensor readings and messages. A 10kΩ potentiometer can be connected to the LCD to adjust its contrast and make the displayed information easier to see (Raj, 2015). In our project, the LCD is used to continuously display the current state of the smoke detection

system, allowing the user to see whether the system is operating normally or an alarm has been triggered.

LEDs

LEDs (Light Emitting Diodes) are the electronic components that produce light when current flows through them, and they are commonly used as visual indicators in electronic systems (bituoelec, 2024). The colour of an LED can be used to represent different conditions or statuses within a system (bituoelec, 2024). LEDs also require a current-limiting resistor to help prevent excessive current from flowing through the LED and the Arduino output pin (Arduino Forum, 2016). In our project, green, yellow and red LEDs are used as visual indicators to show the different states of the smoke detection system.

Piezo Buzzer

A piezo buzzer is an electronic component that produces sound using the piezoelectric effect, where an electrical signal causes a material inside the buzzer to vibrate and produce sound (Kendall, 2023). Piezo buzzers are commonly used to provide audible alerts in electronic systems (Kendall, 2023). In our project, the piezo buzzer is used as an audible alarm when the MQ-2 sensor detects a dangerous level of smoke, which gives the user an additional warning alongside the visual indicators.

Pushbutton

A pushbutton is a switch that allows a user to provide input to an Arduino by opening or closing an electric connection (Arduino, 2024). Arduino explains that a button can be connected to a digital input so that the Arduino can detect when the button is pressed (Arduino, 2024). In our project, the pushbutton is used to reset the alarm after the smoke has been detected which allows the user to stop the alarm once the button is pressed.

Resistors

A resistor is an electric component used to control the amount of current flowing through a circuit. Resistors can be used to protect components by limiting the current supplied to them (Arduino Forum, 2016). In our project, 220 Ω resistors are connected in series with LEDs to limit the current flowing through them. The circuit also includes a 10k Ω , which is used as part of the sensor circuit.

Potentiometer

A potentiometer is a variable resistor whose resistance can be adjusted by turning its control (Raj, 2015). A 10k Ω potentiometer is commonly used with the 16×2 LCD to adjust the contrast of the display (Raj, 2015). In our project, the potentiometer is

connected to the LCD and allows the display contrast to be adjusted so that the information shown on the screen can be viewed clearly.

Capacitor

A capacitor is an electronic component that can store electrical energy and is also used to help smooth changes in a circuit's power supply (Jason, 2024). Capacitors can reduce short-term voltage fluctuations, helping to provide a more stable supply to electronic components (Jason, 2024). In our project, a 100 μF capacitor is included in the circuit to help with power stability.

Breadboard

A breadboard is used to build and test electronic circuits without permanently connecting the components together (Llamas, 2026). The holes on a breadboard are internally connected in groups, allowing components and jumper wires to be connected without soldering (Llamas, 2026). In our project, the breadboard is used to connect the Arduino Uno to the MQ-2 sensor, LCD, LEDs, buzzer and other components while developing and testing the smoke detection system.

Power Supply

The Arduino Uno can be powered through a USB connection or an external power supply (ARDUINO, 2021). When using an external supply, Arduino recommends an input voltage of 7-12V for the board (ARDUINO, 2021). In our project, a 9V battery is used as the external power source for the smoke detection system which provides the power needed for the Arduino and the connected components to operate.

Objectives

- Use an Arduino Uno to control the smoke detection system.
- Use the MQ-2 sensor to detect smoke or gas levels.
- Use the LCD to display the current status of the system.
- Use the green, yellow and red LEDs to show the different levels detected.
- Use the buzzer to alert the user when the detected level becomes too high.
- Use the pushbutton to reset the alarm after it has been activated.
- Build and test the complete circuit in Tinkercad.

Components

Component	Purpose
Arduino Uno	Controls the circuit and processes the readings from the smoke sensor.
MQ-2 Smoke/Gas Sensor	Detects smoke and gas levels in the environment.
16x2 LCD	Displays the status of the smoke detection system to the user.
Green LED	Shows that the system is operating normally.
Yellow LED	Indicates a warning
Red LED	Turns on when a high level of smoke is detected.
Pushbutton	Allows the user to reset the alarm after it has been activated.
Potentiometer	Used to adjust the contrast of the LCD.
Resistors	Limit the current going to the LEDs and are also used where needed in the circuit.
Breadboard	Allows the components to be connected and tested without soldering.
Capacitor	Helps with power stability in the circuit.
9V battery	Provides power to the circuit.

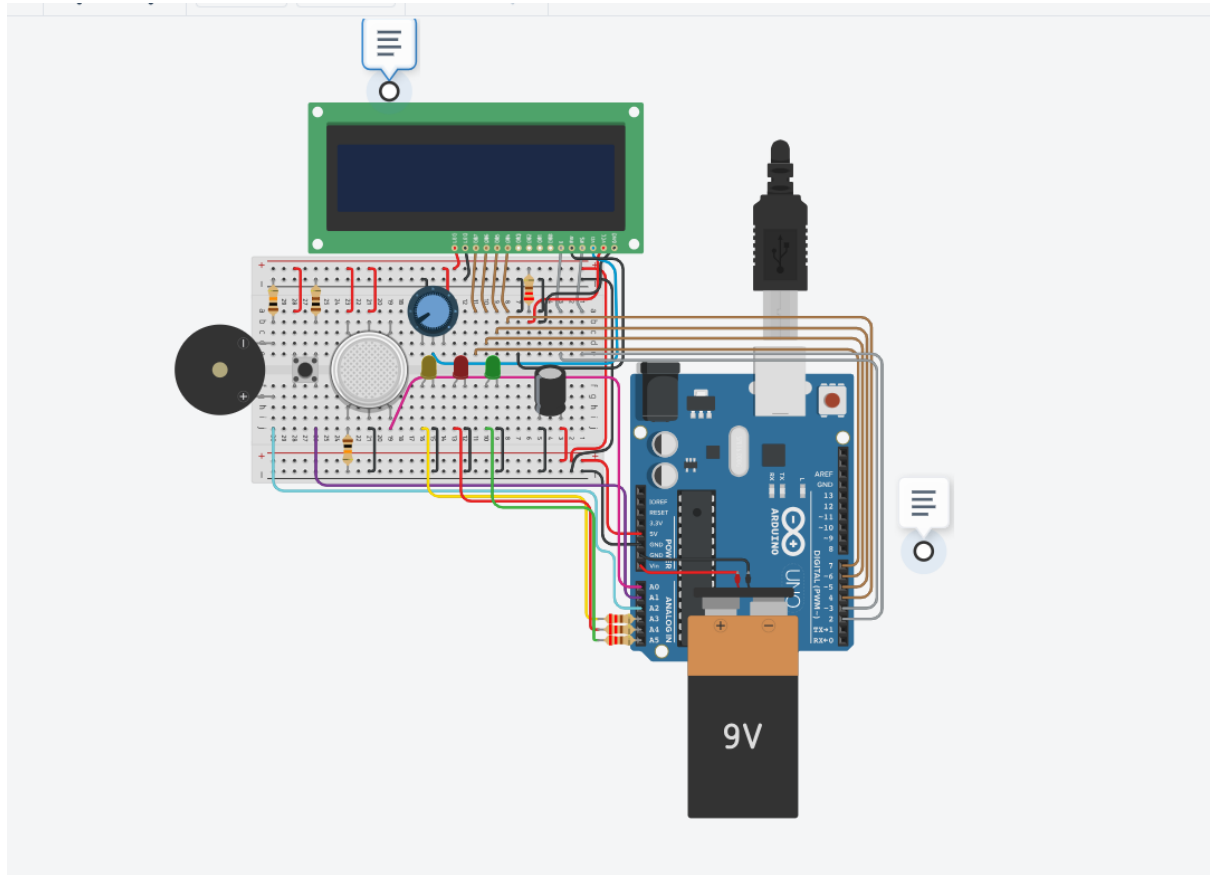
System Design

The smoke detection system is built around the Arduino Uno, which controls the different components in the circuit. The MQ-2 sensor is connected to the Arduino and is used to detect smoke or gas levels. The reading from the sensor is then used by the Arduino to decide what the current status of the system is. The system uses three LEDs to show the different states; the green LED indicates that the system is “safe”, the yellow LED is used to indicate a sensor fault or low power, and the red LED is used when high-level smoke is detected. A piezo buzzer is also connected to the Arduino and is used to provide an audible alarm when the smoke level becomes too high.

The 16x2 LCD is connected to the digital pins of the Arduino and displays information about the system. The LCD uses D2 for RS, D3 for E and D4-D7 for the data pins. A potentiometer is connected to the LCD to adjust its contrast. A pushbutton is also included in the design; it allows the user to rest the alarm after it has been activated. The components are connected using a breadboard, while the 9V battery provides power to the circuit. Resistors are used with the LEDs to limit the current and protect the components.

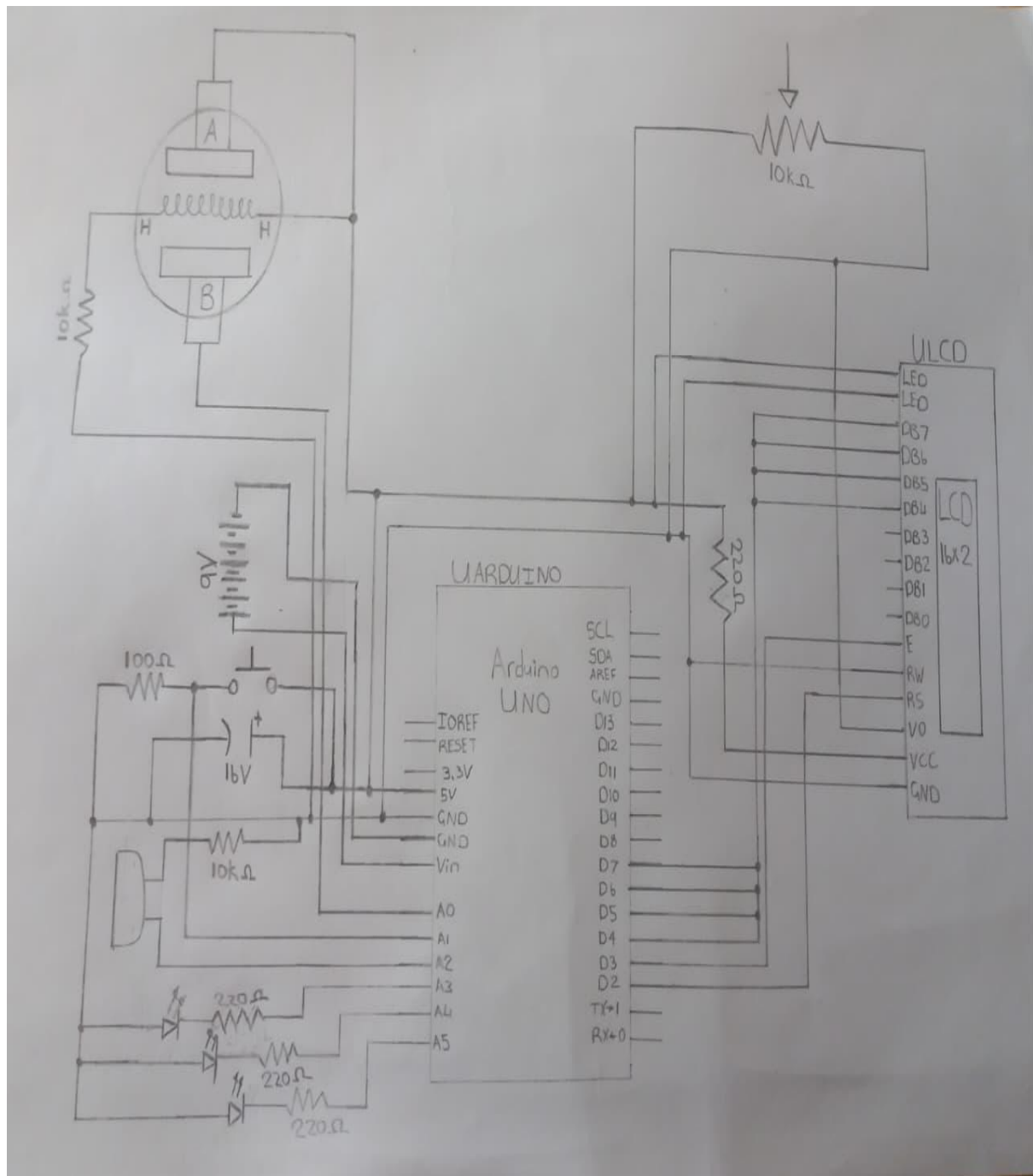
Overall, the system works by detecting smoke → processing the sensor reading with the Arduino → displaying the status on the LCD → using the LEDs and buzzer to alert the user when necessary.

Tinkercad Circuit



Link to Tinkercad circuit: [Circuit design Smoke Detection System - Tinkercad](#)

Circuit Schematic



Arduino Code

```
Text 1 (Arduino Uno R3)
1 #include <LiquidCrystal.h>
2
3 LiquidCrystal lcd_1(2, 3, 4, 5, 6, 7);
4
5 const int gasPin = A0;
6 const int buttonPin = A1;
7 int buttonState;
8 int gas = 0;
9 const int gasMedThreshold = 620;
10 const int gasHighThreshold = 750;
11 const int buzzerPin = A2;
12 int greenLED = A5;
13 int redLED = A4;
14 int yellowLED = A3;
15 bool alarmActive = false;
16 bool ledState = false;
17 const int FLASH_INTERVAL = 300;
18 int lastFlashTime = 0;
19
20 int gasState = 0;
21
22 void setup() {
23   Serial.begin(9600);
24   lcd_1.begin(16, 2);
25   pinMode(buzzerPin, OUTPUT);
26   analogWrite(greenLED, 255);
27
28   lcd_1.print("Gas: Safe");
29 }
30
31 void loop() {
32   buttonState = analogRead(buttonPin);
```

```
Text 1 (Arduino Uno R3)
30
31 void loop() {
32   buttonState = analogRead(buttonPin);
33   gas = analogRead(gasPin);
34
35   int buzzOutput = map(gas, 487, 876, 0, 255);
36   Serial.println(buttonState);
37
38   if (alarmActive && buttonState > 500) {
39     alarmActive = false;
40     analogWrite(greenLED, 255);
41     analogWrite(redLED, 0);
42     analogWrite(yellowLED, 0);
43     noTone(buzzerPin);
44
45     lcd_1.clear();
46     lcd_1.print("Gas: Safe");
47
48     gasState = 0;
49   }
50
51   if (gas > gasHighThreshold && !alarmActive) {
52     if (gasState != 2) {
53       lcd_1.clear();
54       lcd_1.print("Gas: HIGH!");
55       gasState = 2;
56     }
57
58     alarmActive = true;
59     runAlarm();
60   }
61 }
```

Text



1 (Arduino Uno R3)

```
61
62   if (gas > gasMedThreshold && !alarmActive) {
63       analogWrite(greenLED, 0);
64       analogWrite(redLED, 0);
65       analogWrite(yellowLED, 255);
66       tone(buzzerPin, buzzOutput);
67
68       if (gasState != 1) {
69           lcd_1.clear();
70           lcd_1.print("Gas: Medium");
71           gasState = 1;
72       }
73
74   } else if (gas <= gasMedThreshold && !alarmActive) {
75       analogWrite(greenLED, 255);
76       analogWrite(redLED, 0);
77       analogWrite(yellowLED, 0);
78
79       if (gasState != 0) {
80           lcd_1.clear();
81           lcd_1.print("Gas: Safe");
82           gasState = 0;
83       }
84   }
85
86   if (alarmActive) {
87       analogWrite(greenLED, 0);
88       analogWrite(yellowLED, 0);
89
90       lcd_1.setCursor(0, 1);
91       lcd_1.print("!!! ALARM !!!");
92   }
```

Text



1 (Arduino Uno R3)

```
61       lcd_1.print("Gas: Safe");
62       gasState = 0;
63   }
64 }
65
66 if (alarmActive) {
67     analogWrite(greenLED, 0);
68     analogWrite(yellowLED, 0);
69
70     lcd_1.setCursor(0, 1);
71     lcd_1.print("!!! ALARM !!!");
72
73     runAlarm();
74 }
75
76 delay(15);
77 }
78
79 void runAlarm() {
100     if (millis() - lastFlashTime >= FLASH_INTERVAL) {
101         lastFlashTime = millis();
102         ledState = !ledState;
103
104         if (ledState) {
105             analogWrite(redLED, 255);
106             tone(buzzerPin, 300);
107         } else {
108             analogWrite(redLED, 0);
109             noTone(buzzerPin);
110         }
111     }
112 }
```

Code Explanation

The code continuously reads gas concentration from the MQ-2 sensor and writes to a 16x2 LCD and changes three LED indicator to show the system's current state, with a piezo buzzer for audible alerts.

Two thresholds put gas readings into three states: Safe (only the green LED is lit and "Gas: Safe" is displayed), Medium (only the yellow LED is lit and LCD displays a warning), and a buzzer tone whose pitch scales with gas level via `map()`), and High (only the red LED is lit and flashing, continuous buzzer, LCD shows "!!! ALARM !!!"). A `gasState` variable prevents the LCD from refreshing unless the state changes.

During an alarm, the `runAlarm()` function uses a `millis()` function for non-blocking timing (rather than `delay()`) to flash the red LED and pulse the buzzer every 300 ms, keeping the button and sensor responsive while alarmed. The alarm can only be cleared by pressing the pushbutton which resets all outputs and returns the LCD to "Gas: Safe".

The added medium warning tier goes beyond the base requirements, giving users a warning before the full alarm.

Testing and Results

We tested the smoke detection system in Tinkercad by placing the gas cloud over the MQ-2 sensor and checking how the different components responded. When there was no gas cloud, the green LED was on, and the LCD showed safe status. As the gas cloud was placed over the breadboard, the yellow LED turned on, and the LCD displayed the medium warning. When the gas level became high, the red LED started flashing; the buzzer went off and the LCD displayed the alarm message. We also tested the pushbutton while the alarm was active. The alarm stopped for a short moment after pressing the button but then started again because the gas cloud was still over the sensor. Other than the pushbutton issue, the system worked as expected during the testing.

Challenges and Improvements

One of the challenges we had was connecting all components correctly in Tinkercad. There are quite a few components in the circuit; we had to check that they were connected to the correct Arduino pins. We also had to make sure the MQ-2 sensor readings worked properly with the different levels in the code. One improvement would be to make sensor readings more accurate by testing and adjusting the threshold values more carefully.

Conclusion

Overall, we were able to create and test a smoke detection system using an Arduino Uno and the different electronic components. The MQ-2 sensor detects the smoke or gas level, while the LEDs, buzzer and LCD show the current state of the system. The pushbutton allows the alarm to be cleared when needed. Working on this project also helped us understand how the different components are connected and how Arduino code can be used to control them.

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